HARI PRASAD REDDY SHEELAM

Sunnyvale, CA, 94086 | (469) 971-8679 | hariprasadreddysheelam@my.unt.edu | LinkedIn | GitHub

CAREER SUMMARY

Software Engineer with expertise in full-stack development, ML, and cloud solutions. Skilled in deploying Python ML models for predi ctive analytics, anomaly detection, and demand forecasting. Experienced in secure healthcare and enterprise applications using C#, ASP .NET, Angular, Node.js, and SQL/NoSQL databases, with strong proficiency in Azure, AWS, and GCP.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, C#, SQL, C, C++.
- **Data Structures & Algorithms:** Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Tables, Heaps, Sorting & Searching.
- **Web & Frontend Technologies**: HTML5, CSS, React, Redux, Angular 2+, Node.js, Express.js, ASP.NET, J2EE, Spring Boot, XML, JSON, SOAP, MVC & Component-Driven Architecture, System Design.
- **Databases & Management:** SQL Server, MySQL, PostgreSQL, MongoDB, Redis, DB design, normalization, and optimization.
- **Tools and Frameworks:** Gradle, Maven, Docker, Kubernetes, Jenkins, Swagger, Visual Studio Code, Git, GitLab, Apache Kafka, Postman, JIRA, Log4j, Jupyter Notebook, Django, TensorFlow, Grafana**.**
- **Machine Learning & AI:** ML models for predictive modeling, anomaly detection, and demand forecasting; skilled in data preprocessing, feature engineering, model evaluation, and large language models (LLMs) for NLP tasks.
- **Cloud Services:** Azure (IaaS & PaaS, DevOps pipelines), AWS (EC2, S3, Lambda, RDS, DynamoDB), and GCP (BigQuery, Pub/Sub).

PROFESSIONAL EXPERIENCE

Walmart Global Tech | *Software Engineer* **Oct 2024 - Present**

- Engineered advanced ML models in Python, TensorFlow, and PyTorch for inventory anomaly detection (cutting errors by 40%) and predictive demand forecasting (achieving 92% accuracy), driving 35% improvement in order workflow efficiency.
- Established automated, production-grade data ingestion pipelines, reducing data preparation time by 60% and ensuring daily delivery of clean, validated inventory and transaction datasets for analytics and ML pipelines.
- Trained, fine-tuned, and optimized ML models using advanced hyperparameter tuning, boosting anomaly prediction accuracy by 20% and enabling early detection of supply chain risks.
- Designed and implemented highly scalable microservices and optimized GraphQL APIs using Node.js, Express.js, and JavaScript, improving response times by 30%.
- Integrated Fulcrum API data into Test Suite, improving test reliability of healthy item workflows and ensuring 95% data integrity.
- Automated and validated API calls in the Astro ecosystem (create, move, cancel, and return order flows), utilizing asynchronous methods and retry logic, maintaining a consistent 96% pass rate across production and staging.
- Developed and maintained automated 10+ mobile test scripts using Appium and JavaScript, enabling cross-platform iOS and Android testing and improving test execution efficiency during nightly and pre-release cycles
- Developed and maintained automated looper flows using TestCafe, Playwright, Node.js, and TypeScript, boosting test execution efficiency by 50% during nightly and pre-release cycles.
- Designed and implemented a modular web automation framework for the Teflon US market using Playwright and Codegen, reducing manual QA effort by 30% and accelerating regression cycles by 40%.
- Led end-to-end test validation for international releases, including Merge Hallway, CA Teflon, MX Teflon, and MX Bodega Teflon, covering 5+ production environments with 98% accuracy in cross-market functionality.
- Monitored Grafana dashboards for visibility into test health, identifying anomalies that reduced production defects by 25%.
- Coordinated and resolved 10+ P1 incidents as part of the Teflon on-call team, collaborating with IDC teams across time zones and ensuring 100% SLA compliance for critical production issues.
- Managed and contributed to Git repositories, leveraging POM (Page Object Model), async/await, and modular test architecture to improve code reusability by 60% and reduce onboarding time for new team members.
- Actively participated in Agile sprints, contributed to daily stand-ups, backlog grooming, and retrospectives, ensuring continuous delivery and test coverage alignment with evolving product goals.

University of North Texas | *Teaching Assistant* | *Machine Learning* **Aug 2022 - May 2024**

- Mentored 60 students in crafting anomaly detection models, which increased accuracy by 95% using machine learning algorithms, and learned knowledge in Natural Language Processing and Deep Learning.
- Engineered ML models that boosted predictive accuracy by 15%, leveraging NumPy, TensorFlow, scikit-learn, and Pandas.
- Developed and implemented diverse instructional strategies that improved student comprehension by 25%, effectively simplifying complex topics to accommodate varying learning styles.
- Proactively addressed and resolved over 100 programming-related issues for students, enhancing problem-solving skills and fostering a collaborative learning environment.
- Increased programming proficiency in Python, R, and MATLAB by 40%, focusing on machine learning methods, data preprocessing, visualization, and evaluation, resulting in improved student project outcomes.

- Graded and evaluated approximately 200 student submissions, providing actionable feedback that contributed to a 20% improvement in overall student performance in assignments.
- Coordinated with the course instructor and other teaching assistants to streamline instructional materials and grading criteria, facilitating a cohesive educational experience and responding to student inquiries with a 90% satisfaction rate.

Audree Infotech Pvt. Ltd. | *Full-Stack Developer* **Jun 2021 - Jul 2022**

- Delivered critical healthcare web applications for Hetero Drugs using C#, ASP.NET 4.5, and SQL Server, securing PHI and EHR data in compliance with HIPAA, supporting 5+ integrated systems, and significantly improving data integrity and access security.
- Developed ASP.NET Web API v2 endpoints for Hetero Drugs following MVC design patterns, enabling secure integration with healthcare systems and ensuring HL7/FHIR-compliant interoperability for patient and EHR data.
- Implemented asynchronous messaging systems using JMS for Hetero Drugs, enabling non-blocking communication between healthcare services for real-time alerts, patient notifications, and critical medical event processing.
- Configured HIPAA-compliant cross-domain requests for Hetero Drugs via ASP.NET WebApi.Cors NuGet package, ensuring secure data exchange across integrated systems and eliminating potential PHI access violations.
- Developed Angular 2+ front-end applications for Hetero Drugs in Component-Driven Architecture, delivering responsive, user-friendly interfaces that improved clinician workflow efficiency by 30%.
- Automated CI/CD pipelines for Hetero Drugs using Azure DevOps, configuring role-based access control and audit trails for secure, traceable deployments of healthcare applications.

Audree Infotech Pvt. Ltd. | *Jr.Full-Stack Developer* **Jan 2021 - May 2021**

- Designed backend applications within the Java ecosystem, achieving a 90% increase in scalability and response times.
- Employed Spring Boot to deliver Microservices RESTful API endpoints and Log4j to deploy application logging.
- Utilized SQL Server expertise to design and normalize database schemas and reduced data redundancy by 95%.
- Created user-centric web pages using HTML, CSS, and JavaScript to boost the user experience by 90%.

ACADEMIC PROJECTS

Loan Approval Prediction | *University of North Texas* **Jan 2023 - May 2023**

- Deployed a machine learning model using Python to automate decision-making processes for financial institutions.
- Conducted exploratory data analysis and executed multiple classification algorithms using Scikit-learn. Applied Matplotlib and Seaborn for effective data visualization of critical feature impacts on loan approval outcomes.
- Achieved an 82% accuracy rate with the Random Forest Classifier, reducing risks through precise predictions.

Analysis of Uber & Lyft Cab Prices| *University of North Texas* **Jan 2023 - May 2023**

- Analyzed 600K+ ride records using Python and Pandas, achieving 95% data retention post-cleaning and ensuring high-quality, actionable datasets.
- Visualized complex trends & patterns with Matplotlib & Seaborn, uncovering critical insights for ride-sharing strategy optimization.
- Performed rigorous statistical hypothesis testing with SciPy & StatsModels, revealing a 12–18% cost advantage for Uber ($p < 0.05$) and informing data-driven pricing decisions.

Custom Shell | *University of North Texas* **Aug 2022 - Dec 2022**

- Built a custom shell using C and Golang, integrating command history, parsing, and execution of system commands.
- Executed command parsing logic to separate user input into commands and arguments, implemented a history management feature for retrieval of old commands, and ensured accurate execution of built-in and external commands.
- Produced 95% operational effectiveness in managing file operations and complex command sequences.

Scheduling Algorithms | *University of North Texas* **Sep 2022 - Oct 2022**

- Developed a C++ simulation platform in a Linux environment to analyze over 6 CPU scheduling algorithms, including FCFS, SJF, Round Robin, Priority, and Multi-Level Queue Scheduling.
- Evaluated over 1,000+ process scenarios, measuring performance metrics like average waiting time, turnaround time, and CPU utilization, resulting in a 40% improvement in comparative scheduling accuracy.
- Delivered data-driven insights that enhanced the algorithm, optimizing system throughput and responsiveness by 35%.

Eagle Business Platform | *University of North Texas* **Sep 2022 - Dec 2022**

- Constructed a platform, a robust web-based application that empowers users to compare and invest in automobiles.
- Expertly managed databases using MySQL Workbench and Microsoft Visual Studio, ensuring efficient data handling and optimization. Enhanced backend functionality with Node.js and generated a responsive design with React.
- Secured over 300 user registrations in the first month and enhanced the effectiveness of investment analysis by 90%.

EDUCATION

University of North Texas **Aug 2022 - May 2024**
Master of Science, Computer Science Engineering

- **Achievements:** ABET Accredited
- **Coursework:** Machine Learning, Software Development for AI, Fundamentals of AI, Big Data, Computer Algorithms.

CERTIFICATIONS

- IBM Full Stack Software Developer Professional, AWS Solution Architect Associate. **Jan 2024 - May 2024**